TERBUTALINE

Not for prolonged tocolysis

Tremor

CHEMICAL NAME (RS)-5-[2-(tert-butylamino)-1-hydroxyethyl]benzene-1,3-diol (sulfate)
CLASS Short-acting β2-adrenoceptor agonist
TRADE NAMES Bricanyl · Brethine · Bricanyl Turbohaler

THERAPEUTIC USES / INDICATIONS

- Relief of bronchospasm in asthma and COPD.
- Acute severe asthma (SC/IV).
- Uterine hyperstimulation / short-term tocolysis (e.g. during transfer or fetal distress).

PHARMACODYNAMICS

- **β2 agonist** → bronchial and uterine smooth muscle relaxation.
- Some β1 effects at higher doses.

ADMINISTRATION

- **Inhaled** (dry-powder Turbohaler).
- **SC** injection; **IV** infusion.
- **PO** tablets/syrup (rarely).
- Tocolysis: single SC dose only (not prolonged use).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Tremor, nervousness.
- Tachycardia, palpitations.
- Headache, cramps.

ADVERSE EFFECTS (SERIOUS)

- Hypokalaemia; hyperglycaemia.
- Maternal tachycardia, arrhythmias, pulmonary oedema, myocardial ischaemia (prolonged tocolysis); hence warning against >48–72 h use.
- Paradoxical bronchospasm.

DOSAGES

- **Inhaled:** 500 mcg (1 inhalation) as needed, up to 4 times daily.
- **SC (asthma):** 250–500 mcg up to 4 times daily.
- **Tocolysis / uterine hyperstimulation:** 250 mcg SC once (may repeat once).
- **PO:** 2.5–5 mg three times daily.

CONTRAINDICATIONS & CAUTIONS

- Prolonged tocolysis (>48–72 h) or oral tocolysis.
- Cardiac disease in pregnancy; pre-eclampsia; antepartum haemorrhage (for tocolysis).
- **Caution:** hyperthyroidism, diabetes, arrhythmias.

KEY INTERACTIONS

- **β-blockers:** antagonism.
- **Corticosteroids, diuretics:** hypokalaemia.
- **Halothane:** arrhythmias.

PHARMACOKINETICS

- **Onset:** inhaled 5 min; SC 5–15 min. **Duration:** 4–6 h.
- Metabolised in liver/gut wall.
- Excreted in urine. **t½:** 3–4 h (terminal ~16 h).

NURSING MANAGEMENT

- Maternal HR, BP, SpO_2; fetal heart monitoring.
- Hold and report if maternal HR >120 or chest pain/breathlessness.
- Monitor K^+ and glucose.
- Respiratory assessment in asthma.

PATIENT EDUCATION

- Inhaler technique for Turbohaler (no shaking; strong, deep breath).
- Report chest pain, breathlessness or palpitations.

KEY POINTS

- SABA similar to salbutamol.
- Not for prolonged tocolysis.
- Single 250 mcg SC for uterine hyperstimulation.
- Tremor, tachycardia, hypokalaemia.

SOURCES CHECKED **[449]** Bricanyl Respules 2.5 mg/ml Nebuliser Solution (UK SmPC) **[450]** Bricanyl Injection, 0.5 mg/ml, solution for injection or infusion (UK SmPC) **[451]** Terbutaline Sulfate, oral (US label) **[452]** Terbutaline Sulfate, oral (US label, Chartwell RX)
Full references p. 432–457